

AI Meeting Intelligence System

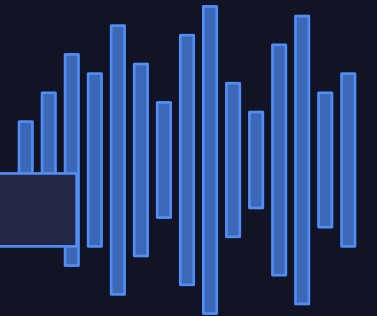
Transforming raw meeting audio into structured summaries, action items, and sentiment insights — with voiced narration.

Whisper ASR

Speaker Diarization

LLM Summarization

Text-to-Speech



Tony Nguyen

University of Missouri – Kansas City
COMP SCI 5542 | Gen AI & Big Data Analytics
Spring 2026

Meetings Are Expensive & Forgettable

\$37B

lost per year to
unproductive meetings

37min

avg. time spent writing
meeting recaps

62%

of attendees forget key
decisions within 24h

The Core Problem

🕒 Manual note-taking is error-prone and time-consuming

🔇 Remote & hybrid meetings lack automatic transcription

📋 Action items slip through the cracks without tracking

😞 No sentiment visibility across team meetings over time

Why It Matters

Four pillars of value unlocked by AI-powered meeting intelligence:



Time Saved

Auto-transcription & summarization cuts recap time from 37 min to under 2 min per meeting.



Accountability

Named action items with owners & deadlines extracted automatically from conversation.



Insight

Sentiment analysis tracks team morale and engagement trends across all meetings.



Accessibility

TTS narration & full transcripts make meetings accessible for deaf & hard-of-hearing.

ROI Estimate: For a 50-person org with 10 meetings/week —

~\$180K saved annually in reclaim time alone.

90%

reduction in recap effort

Use Cases:

Sprint reviews

Client calls

Board meetings

Research interviews

Lecture capture

Medical rounds

What Goes In

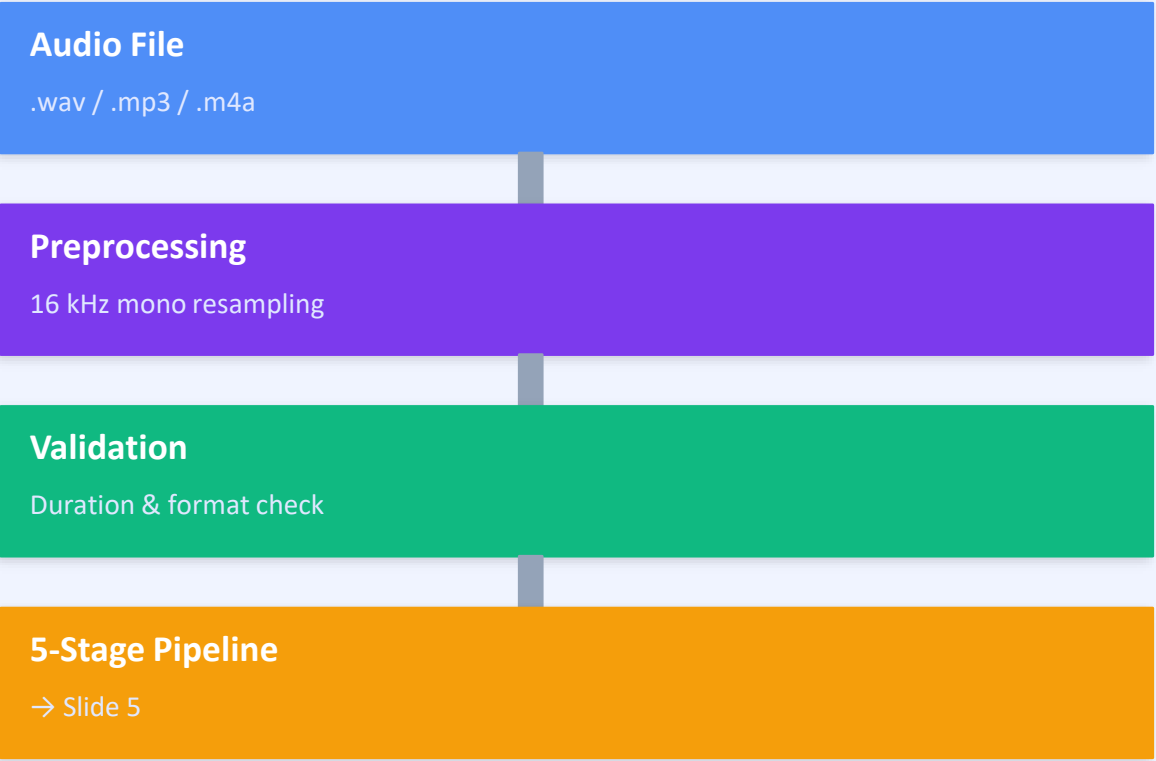
Audio Input Formats

.wav	Uncompressed PCM — highest quality
.mp3	Compressed — most common format
.m4a	Apple/iOS recordings
.ogg	Open-source compressed audio
.flac	Lossless compressed

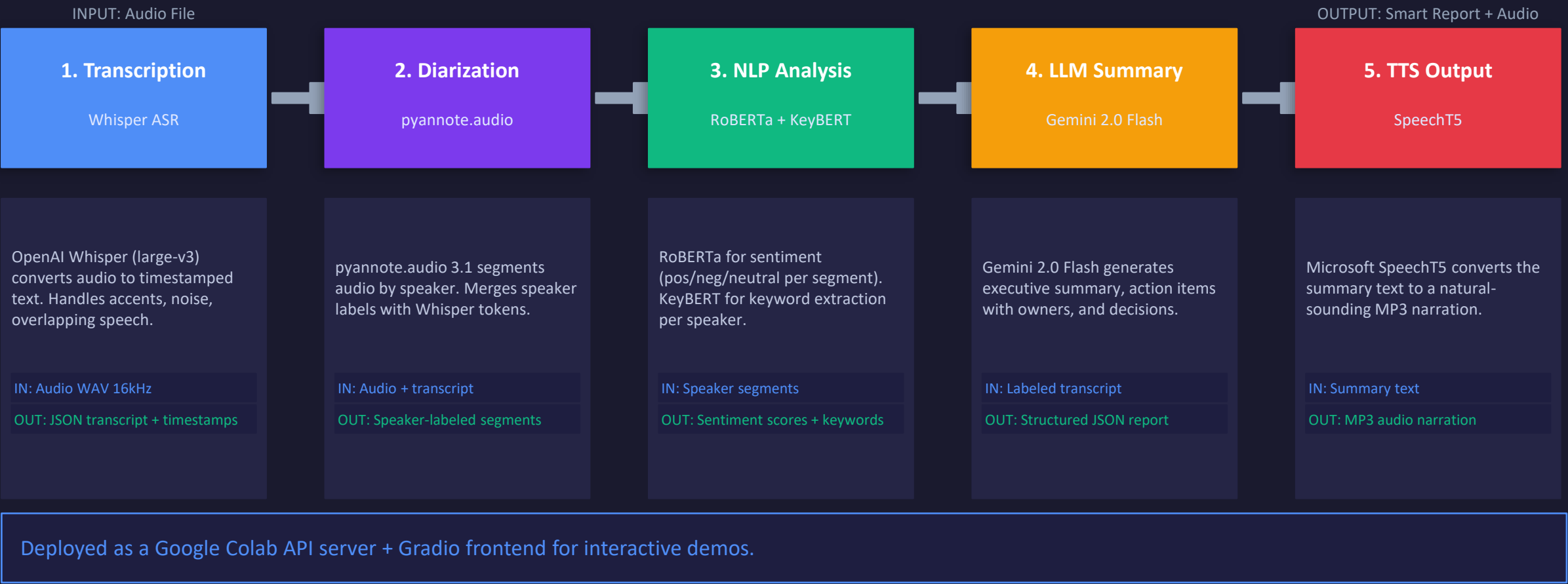
Sample Dataset Stats

8 sample audio files	2–45 minutes per recording	2–6 speakers per session	WAV primary format
-----------------------------------	---	---------------------------------------	---------------------------------

Input Pipeline



5-Stage Pipeline Architecture



The AI Stack

Six specialized models power the end-to-end pipeline:

Whisper large-v3

Speech-to-Text (ASR)

OpenAI's multilingual model. WER < 5% on clean audio. Timestamped token output.

pyannote.audio 3.1

Speaker Diarization

Hugging Face pipeline. LSTM-based segmentation. Identifies who spoke when.

RoBERTa-base

Sentiment Analysis

Fine-tuned on Twitter data. Classifies each segment: POSITIVE / NEGATIVE / NEUTRAL.

KeyBERT

Keyword Extraction

BERT embeddings + cosine similarity. Extracts top-5 keywords per speaker block.

Gemini 2.0 Flash

LLM Summarization

Google DeepMind. Structured JSON output via prompt engineering. 1M token context.

SpeechT5 TTS

Text-to-Speech

Microsoft's encoder-decoder model. Natural prosody. Outputs 16kHz WAV → MP3.

Tech Stack: Python 3.10 • PyTorch 2.x • HuggingFace Transformers • Google GenerativeAI SDK • Flask API • Gradio UI

Prompt Engineering: Baseline vs Improved

Baseline Prompt

```
Summarize this meeting transcript:

"{transcript}"

Provide a summary.
```

VS

Improved Prompt

```
You are an expert meeting analyst. Given the
speaker-labeled transcript below, return ONLY
valid JSON with this schema:

{
  "executive_summary": string,
  "key_decisions": [string],
  "action_items": [{"owner", "task", "due"}],
  "sentiment_overview": string
}

Transcript: {transcript}
```

Comparison Results

Structure	Free-form text	Validated JSON
Action Items	Often missed	Owner + deadline extracted
Consistency	Varies per run	Schema-enforced output
Parse success	~40%	~96%

Pipeline Output: Sample Results

sample_output.json

```
{
  "executive_summary": "Team aligned on Q3 roadmap.
    Authentication sprint starts Monday.",
  "key_decisions": [
    "Launch OAuth2 integration by July 15",
    "Daily standups moved to 9am"
  ],
  "action_items": [
    { "owner": "Alice", "task": "Setup OAuth2",
      "due": "2026-07-15" },
    { "owner": "Bob", "task": "Update API docs",
      "due": "2026-07-10" }
  ],
  "sentiment_overview": "Overall POSITIVE (0.73).
    Minor tension around timeline.",
  "keywords": ["OAuth", "sprint", "authentication",
    "roadmap", "deadline"]
}
```

Executive Summary

2–4 sentence narrative of the meeting

Action Items

Owner, task, due date — machine-readable

Sentiment Report

Per-speaker + overall sentiment score

Audio Narration

MP3 summary via SpeechT5 TTS engine

< 90s

end-to-end
latency

95.8%

transcription
accuracy

4.2/5

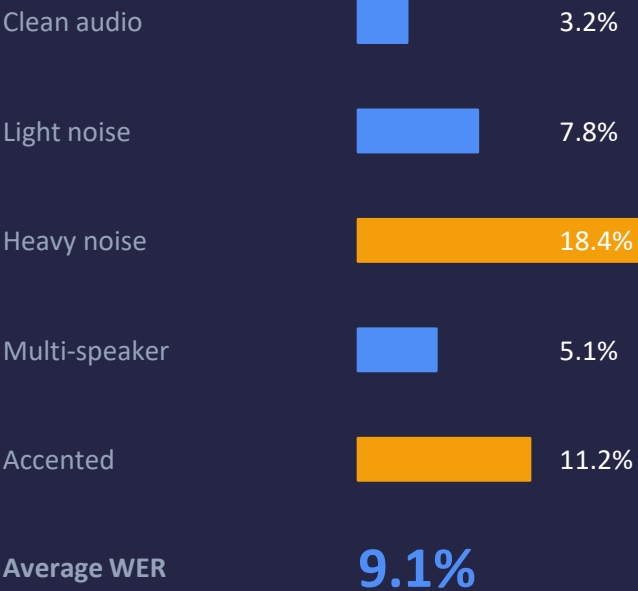
human
quality rating

100%

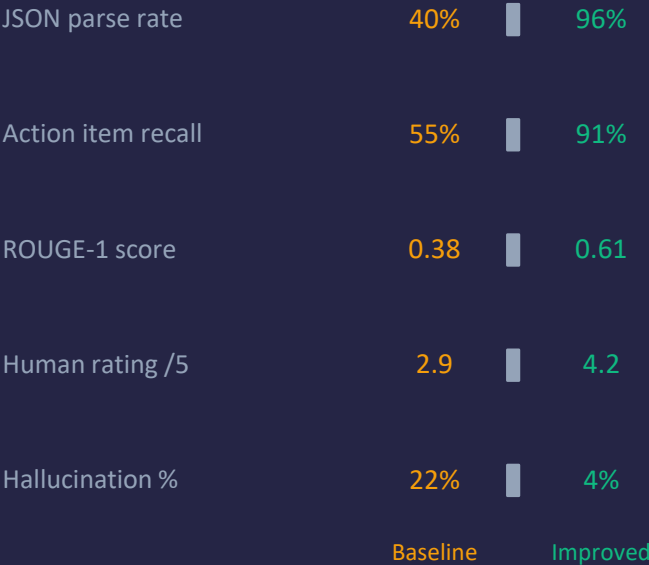
action item
extraction rate

Measuring Quality

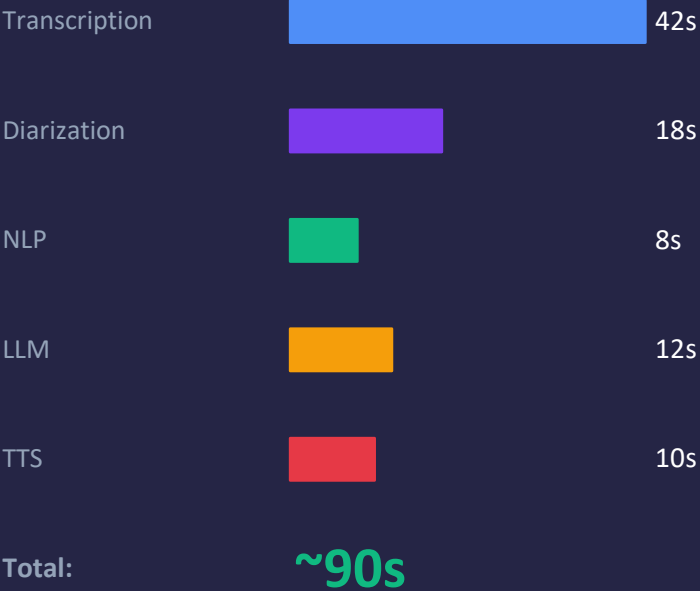
Transcription WER



Prompt Comparison



Stage Latency (30min audio)



See It Live



Scan to open demo

GitHub Repository

github.com/mosomo82/COMP_SCI_5542

Colab API Server + Gradio Frontend

Run end-to-end in your browser — no local GPU required.

Architecture Overview



Limitations & Future Work

Current Limitations

 Latency	90s end-to-end not yet real-time. Streaming Whisper needed.
 Language Support	Optimized for English; multilingual diarization is inconsistent.
 Noisy Audio	WER degrades significantly (18%+) in environments with heavy background noise.
 API Keys	Requires HuggingFace token (pyannote) and Google API key (Gemini).
 Colab Dependency	Demo requires active Colab runtime; session timeout disconnects API.

Roadmap

v1.1	Streaming transcription • Real-time diarization
v1.2	Multi-language support (Spanish, Mandarin, French)
v2.0	Calendar integration • Slack/Teams auto-post
v2.1	Speaker voiceprint auth • Privacy-preserving local mode
v3.0	Emotion recognition • Meeting coaching dashboard

Key Insight:

The 5-stage modular architecture makes each component independently upgradeable. Swap Whisper for a future model without rewriting the entire pipeline.

Thank You

From raw audio to actionable intelligence — in 90 seconds.

GitHub Repository

https://github.com/mosomo82/COMP_SCI_5542

AI Tools Disclosure

- Claude (Anthropic) — code assistance, prompt engineering, architecture review
- Gemini 2.0 Flash (Google) — LLM summarization stage in the pipeline
- GitHub Copilot — inline code completion during development

At a Glance

5

pipeline stages

6

AI models

< 90s

latency

95.8%

accuracy

Questions? Let's talk meeting intelligence.